Electronic supplementary material

Interface removal reshapes the predicted homodimers of human splice isoforms while added sequence seldom builds interfaces

Taner Karagöl and Alper Karagöl

Numbers in square brackets refer to the reference list of the main article.

Supplementary methods

Isoform sequences and pairs

Human UniProtKB/Swiss-Prot entries and their isoform sequences were retrieved from the UniProt REST API on 7 October 2026 (release 2026_03 of 2 September 2026; query reviewed:true AND organism_id:9606, with isoforms included), together with each entry's alternative products and SUBUNIT comment [3]. Every isoform record other than the displayed canonical sequence was paired with the canonical sequence of its entry, which gave 22,131 pairs. Pairs were analysed when both sequences were between 15 and 1,500 residues long (20,171 pairs).

Classification of isoforms against the canonical sequence

Shared sequence was defined by the identical blocks of at least five residues that the SequenceMatcher class of Python's difflib module finds between canonical and isoform with its automatic junk heuristic switched off [55]. The blocks are co-linear and do not overlap, so each canonical residue maps to at most one isoform residue. Isoform residues outside the blocks were counted as added and canonical residues outside the blocks as removed, and each was assigned to the N-terminus, the C-terminus or the interior by its position relative to the first and last block. A pair was a terminal truncation when residues were removed only at one or both termini and none were added, an internal deletion when at least one interior segment was removed and none were added, an addition when residues were added and none were removed, and a replacement when residues were both removed and added. The 8 pairs with no shared block were set aside. For the transition analyses, terminal truncations and internal deletions were pooled as removal-only isoforms.

Homodimer predictions and the high-confidence call

For each of the 32,719 canonical and isoform accessions of the human entries that have isoforms, the complex endpoint of the AlphaFold Protein Structure Database API was queried on 7 October 2026 [34, 35]. A served homodimer was recorded when a returned entry had assembly type Homo, oligomeric state dimer and the accession as its only component with stoichiometry two; 2,190 accessions had one. Model files (mmCIF, version 1) were downloaded for every served homodimer in the analysed pairs. Monomer summaries (mean pLDDT and the fractions of residues

in each pLDDT band) were taken from the prediction endpoint for the accessions with a monomer model (version 6).

The release's manifest (model_entity_metadata_mapping.csv, 4.58 GB, 19,129,717 rows, one homodimer model and one accession per row) lists the homodimer predictions with their ipTM, the ipSAE, pDockQ2 and LIS values computed in each chain direction, and clash counts [35]. Its human rows (91,641, one per accession) and the rows of the identical-sequence copies described below were extracted with Polars on Google Colab. Every homodimer served for the accessions in our pairs had a manifest row under its accession. For each value computed in two chain directions we used the larger, which agrees within 0.01 with the two-decimal ipSAE and pDockQ2 that the API reports for all 2,190 served homodimers. Deposition was reproduced exactly by ipSAE of at least 0.6 and pDockQ2 of at least 0.23 for all 29,447 queried accessions with a manifest row, and a homodimer is called high confidence in this work on that basis. Predicted chain lengths ranged from 30 to 1,500 residues. Transitions were analysed for the 18,508 pairs in which both accessions had a manifest row and each model's chain length equalled the length of the current UniProt sequence, a check that excluded 18 pairs involving 8 canonical and 2 isoform models of earlier sequence versions; the isoform gained or lost the call when its status differed from the canonical's.

Structural features and interface retention

Models were parsed with gemmi [56] after removal of hydrogens. A residue was assigned to the interface when any of its heavy atoms lay within 5.0 Å of a heavy atom of the other chain, found with a k-d tree in SciPy [57]. Per-residue pLDDT was read from the B-factor column, and the lower of the two chains' values was used in residue-level analyses. For the 726 isoform homodimers that contain added residues, the solvent-accessible surface area of each residue of chain A was computed in the isolated chain with the Shrake-Rupley algorithm in Biopython (100 points per atom) [58, 59] and converted to relative accessibility with the theoretical maxima of Tien and colleagues [60]; residues with relative accessibility of at least 0.25, the threshold that best separates surface from interior residues [61], were treated as exposed. Added residues of an isoform model were located by comparing the model sequence with the canonical sequence by the block method above.

For each pair with a high-confidence canonical homodimer whose model sequence matched the UniProt canonical sequence (1,678 of 1,681; 1,662 in the analysed pairs), the canonical interface was the union of the interface residues of the two chains of the canonical model, and the fraction of those residues lying in blocks shared with the isoform was recorded. The fraction of confidently predicted canonical residues (pLDDT of at least 70) that the isoform removes was recorded in the same way. For addition-only isoforms, an added segment counted as inserted next to confidently predicted canonical residues when the canonical residues on both sides of the insertion point, or the single adjacent residue of a terminal extension, had pLDDT of at least 70. When both homodimers of an analysed pair were high confidence (871 pairs, in all of which the

canonical model's sequence matched the current UniProt sequence), the canonical interface residues present in the isoform were compared with the isoform's interface residues that map to the canonical, and their overlap was expressed as a Jaccard index.

Monomer confidence, swaps and robustness

Per-residue pLDDT of the AlphaFold monomer models (version 6), which the database now provides for isoforms [62], was fetched for every isoform and canonical accession of the classified pairs from the model confidence files of the AlphaFold Database, and kept when the model sequence was the current UniProt sequence, verified by the MD5 of the sequence. A residue was added or removed as defined above, and its pLDDT read from the isoform or canonical monomer model; the 376,017 added residues analysed are those of the 12,216 isoforms with both added sequence and a monomer model. For each replacement, every gap between shared blocks in which the isoform both removed and added residues was a swap. The removed and added segments of each swap were aligned locally [63] with the Biopython PairwiseAligner (BLOSUM62 [64], gap opening 11, extension 1) [59], and the alignment score was compared with 100 alignments of the shuffled added segment to give a z-score [65]; a swap was homologous when the z-score was at least 5, at least 20 residues aligned and the aligned identity was at least 0.25.

To test whether isoforms remove interface residues preferentially, each canonical residue of the 1,533 high-confidence pairs whose isoform removes residues was labelled interface or not, placed in a pLDDT band (below 50, 50 to 70, 70 to 90, 90 and above) by the lower of the two chains' values, and marked removed or kept. The odds ratio of removal for interface against other residues was pooled across bands by the Mantel-Haenszel estimator [66], with a 95% interval from 1,000 bootstrap resamples over canonical entries. For the losses that keep the whole interface, the residues an isoform shares with its canonical were compared between the two monomer models, and the mean difference in pLDDT (isoform minus canonical) summarized the confidence of the shared fold.

Transitions were also computed under alternative calls: the ipSAE threshold alone, the pDockQ2 threshold alone, an ipTM of at least 0.8 and of at least 0.6, and a grid of ipSAE thresholds (0.5, 0.55, 0.6, 0.65, 0.7) with pDockQ2 thresholds (0.15, 0.23, 0.3). The agreement of ipTM bands with the deposition call was taken over all queried human accessions with a manifest row. Each isoform was assigned the mechanism that produces it from the UniProt alternative-products annotation, as alternative splicing, alternative initiation, an alternative promoter or a combination of these.

Disorder was also predicted from each isoform sequence with AIUPred [46], which takes a single sequence and no alignment, and residues with a score of at least 0.5 were called disordered; in the residue-level interface model, AIUPred disorder in four bands (below 0.25, 0.25 to 0.5, 0.5 to 0.75, 0.75 and above) replaced the pLDDT band or was added to it. The MANE Select transcript of each entry [51] was read from the UniProtKB cross-references of the same release, and the

transitions and loss models were repeated over the pairs whose canonical sequence is the MANE Select product. For the four isoforms whose added residues form a high-confidence interface, the peptides that UniProtKB maps to the isoform sequence were retrieved through the Proteins API, and a PeptideAtlas [50] peptide was counted as isoform-specific when it covered at least one added residue and occurred neither in the canonical nor in any other human Swiss-Prot gene of the release; the protein sequences of the gene in Ensembl [52] and RefSeq were compared with the isoform and the canonical by identical blocks of at least five residues.

Identical sequences and annotation

To test whether predictions repeat for identical input, 150 canonical entries with and 150 without a high-confidence homodimer were drawn at random (seed 7) from the canonical entries of the classified pairs; 15 of the latter had no predicted homodimer. For each, the UniParc entry of its sequence was retrieved through the UniProt REST API, and every active UniProtKB accession of another entry attached to it was recorded as an identical copy (1,461 copies of 108 human sequences, from 258 species including 44 other human entries). Copies were queried at the complex endpoint and looked up in the manifest as above, and for every copy served as a homodimer the full API entry, with its two-decimal ipSAE and pDockQ2, was retrieved as well.

The SUBUNIT comment of each canonical entry was used after removing statements that UniProt attributes to specific isoforms. Entries were classed as homodimer when the text mentions a homodimer, as other homo-oligomer when it mentions another homo-oligomeric state and no homodimer, as monomer when it states that the protein is a monomer and mentions no homo-oligomer, as other text when it has subunit text of any other kind, and as no text otherwise.

Statistical analysis

Proportions are given with Wilson score 95% confidence intervals [67], and two-by-two comparisons used two-sided Fisher's exact tests, without correction for multiple comparisons. Logistic regressions were fitted by maximum likelihood, and odds ratios are reported with Wald 95% confidence intervals. The gain model was fitted to the analysed pairs whose canonical homodimer was not high confidence and included class, the base-10 logarithm of canonical length, the canonical's fraction of residues with monomer pLDDT below 50 (per 10 percentage points) and the difference in mean monomer pLDDT between isoform and canonical. The canonical's fraction of residues with pLDDT below 50 is the pLDDT-based measure of disorder used in our analysis of truncated isoforms [42]. The loss models were fitted to the analysed pairs whose canonical homodimer was high confidence, with class and, successively, the percentages of the canonical interface and of the confident canonical residues removed (per 10 percentage points), and in the full model the monomer covariates above and the logarithm of canonical length. A separate loss model for the 667 pairs that keep every canonical interface residue included class, the monomer covariates and the logarithm of canonical length. The residue-level model compared exposed added and shared residues of the 726 isoform models without and with pLDDT band (below 50, 50 to 70, 70 to 90, 90 and above) as a covariate. Correlations are Spearman's rank correlations.

Data and software versions

UniProtKB release 2026_03; AlphaFold Database complex models version 1 and monomer models version 6, and the release manifest, accessed 7 October 2026. Python 3.13.16 with pandas 3.0.5, NumPy 2.5.3 [68], SciPy 1.18.1, statsmodels 0.15.0, gemmi 0.7.5 and Biopython 1.88; figures were drawn with Matplotlib 3.10.9 and the workbook of Supplementary Tables S1 to S9 written with openpyxl 3.1.5, both under Python 3.10.12; the manifest was read with Polars 1.35.2, and disorder was predicted with AIUPred 3.1.3 under PyTorch 2.14.1.

Supplementary results

Pairs set aside from the analysis

Of the 20,163 classified pairs, we therefore analysed the 18,508 in which both homodimers were predicted from sequences of the current UniProt length, and set aside 1,655. In 1,381 of the pairs set aside the canonical homodimer was not predicted, in 256 only the isoform homodimer was missing, and in 18 a model's chain length differed from the current UniProt sequence. Of the 679 isoforms without a prediction, 457 were longer than 1,000 residues, and scoring pairs from the served models alone would have counted the 18 pairs with a high-confidence canonical and no isoform prediction as losses.

Reproducibility for identical sequences

Interface scores repeat closely for identical input. We drew 150 canonical entries with and 150 without a high-confidence homodimer and found, through UniParc, 1,461 other UniProtKB entries with identical sequences for 108 of them, from 258 species (Methods). At least 625 of these copies had been predicted, for 79 of the human sequences. Of these, 595 have a manifest row under their own accession, and 30 copies of 13 human sequences are served by the API as deposited homodimers without one. For the 595, the ipSAE differed from that of the human entry by a median of 0.00, by at most 0.02 in 89.2% and by at most 0.05 in 99.0% of pairs (Spearman's ρ = 0.99; Supplementary Figure S1a), and 71 of 76 copies of human sequences with ipSAE between 0.5 and 0.7 fell on the same side of 0.6. The full call agreed for 568 of these 595 copies; the 27 disagreements came from 4 of the 79 human sequences, and 22 of them arose from pDockQ2 alone, which differed between identical sequences by up to 0.76 (Supplementary Figure S1b). The two thresholds reproduced deposition for all 595. The 30 copies without a manifest row were deposited, as were all 13 of their human sequences, and their two-decimal API ipSAE lay within 0.01 of the human entry's manifest value in all 30 and their pDockQ2 in 26.

Calibration against UniProt annotation

The call agrees with curated annotation but recovers a minority of it. Of the canonical entries with a homodimer predicted from their current sequence whose UniProt SUBUNIT comment describes a homodimer, 394 of 1,259 (31.3%) had a high-confidence homodimer, as did 93 of 477 (19.5%)

described as other homo-oligomers, against 12 of 259 (4.6%) described as monomers, 232 of 4,498 (5.2%) with other subunit text and 200 of 2,807 (7.1%) without subunit text (Supplementary Figure S1c). When both the canonical and the isoform homodimer were high confidence, they used the same interface, with a median Jaccard index of 0.96 for addition-only, 0.92 for removal-only and 0.92 for replacement pairs between the canonical interface residues present in the isoform and the isoform's interface residues, and 20 of 871 pairs fell below 0.5 (Supplementary Figure S1d).

Alternative definitions of the call

The findings hold under alternative definitions of the call. Loss rose with interface removal, and in the class order addition, removal, replacement, under the ipSAE threshold alone, the pDockQ2 threshold alone, an ipTM cutoff of 0.8 or of 0.6, and a grid of ipSAE thresholds from 0.5 to 0.7 with pDockQ2 thresholds from 0.15 to 0.3 (Supplementary Figure S1e and S1f, Supplementary Table S7). On the ipSAE threshold alone, the reading we recommend for gains, isoforms that kept every interface residue lost the call in 9.6% of pairs and those that removed more than half in 93.3%, the same pattern at lower absolute rates than under the deposition call.

Disorder predicted from sequence at the interface

Among the 196,198 exposed residues of the 726 high-confidence isoform homodimers that contain added residues, the odds ratio for an added residue to lie at the interface was 0.24 (0.22 to 0.25) without covariates, 0.95 (0.88 to 1.03) with pLDDT band as the covariate, 0.34 (0.32 to 0.37) with AIUPred disorder band in its place and 0.94 (0.87 to 1.01) with both.

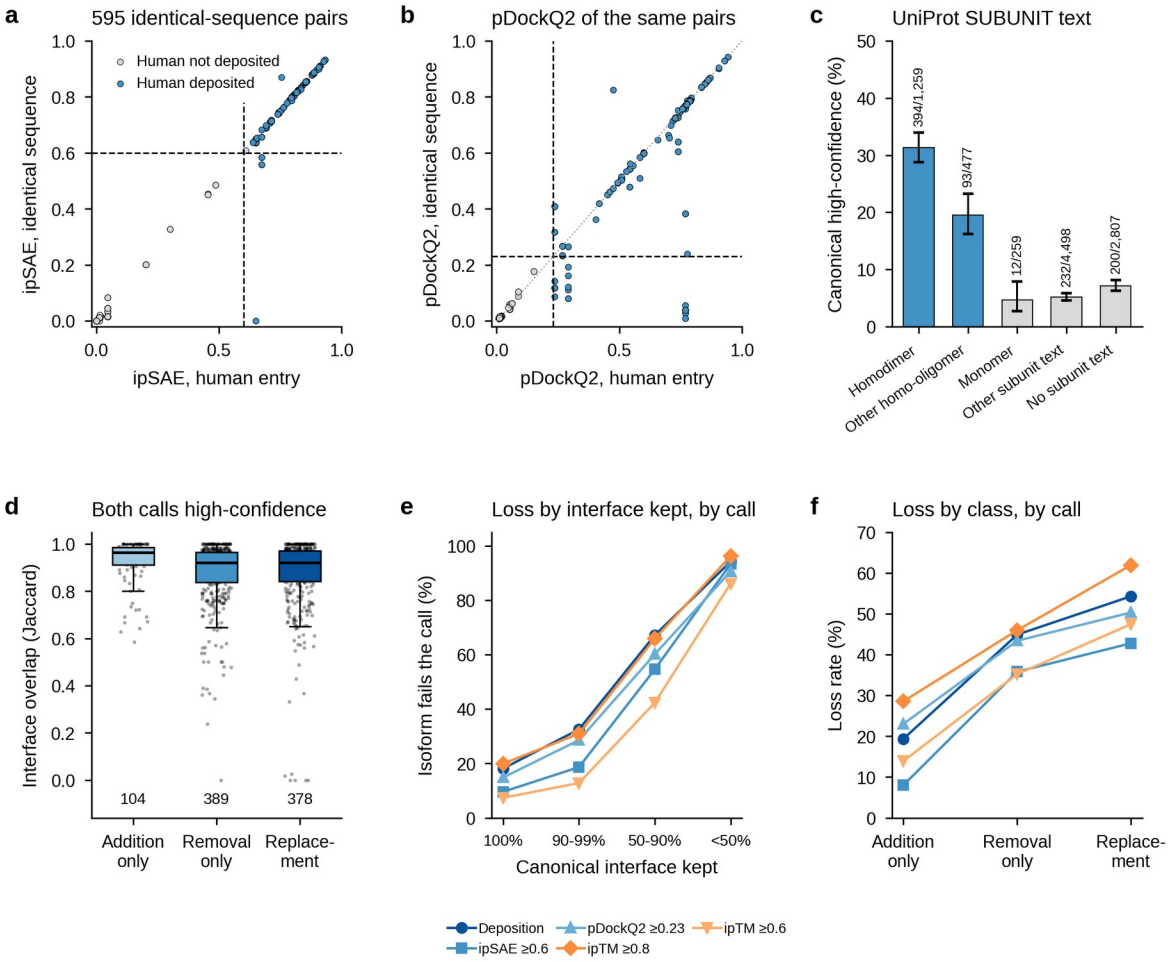
Protein evidence for the candidate isoforms

The added sequence of all four is detected at the protein level, with 21, 4, 88 and 19 PeptideAtlas [50] peptides that cover added residues of the MROH8, CTAG2, ZNF451 and SNX18 isoforms and occur neither in their canonicals nor in any other human Swiss-Prot gene (Table 2). Isoform 2 is the MANE Select product [51] of both CTAG2 and SNX18, so in these genes UniProt's canonical is the alternative form, and all but one of the 447 residues of the added segment of MROH8 isoform 2 lie within the longest MROH8 proteins of Ensembl [52] and RefSeq, of 1,042 and 1,052 residues, neither of which matches a UniProt MROH8 sequence. Isoform 4 of ZNF451, whose canonical is the MANE Select product, is the one minor isoform among the four and the clearest prediction of a homodimer interface built from sequence absent from its gene's principal product.

The MANE Select product as the reference

The results also hold with the principal transcript product as the reference. UniProt's canonical is the MANE Select product in 15,060 of the 18,508 analysed pairs, and among them the class odds ratios for loss of 4.09 and 5.79 fell to between 1.11 and 1.38 once interface removal was in the model, with every 95% interval spanning 1, and isoforms that kept every interface residue lost the call in 17.9% of pairs and those that removed more than half in 94.3%.

Supplementary figure



Supplementary Figure S1. The call is reproducible for identical input and robust to the thresholds.

(a) ipSAE of human entries and of identical sequences in other UniProtKB entries, for the 595 copies with a manifest row, coloured by whether the human homodimer was deposited; dashed lines mark 0.6. (b) pDockQ2 of the same pairs; dashed lines mark 0.23 and the dotted line identity. (c) Canonical entries with a homodimer predicted from their current sequence (9,300) that have a high-confidence homodimer, by the content of their UniProt SUBUNIT comment, with Wilson 95% confidence intervals. (d) Overlap (Jaccard index) of canonical and isoform interface residues for the 871 pairs in which both homodimers are high confidence and the canonical model matches the current UniProt sequence, restricted to residues present in both sequences. (e) Percentage of pairs in which the isoform fails the call, by the fraction of canonical interface residues kept, under the deposition call, the ipSAE threshold alone, the pDockQ2 threshold alone and ipTM cutoffs of 0.8 and 0.6. (f) Loss rate by isoform class under the same five calls.

Supplementary tables

Supplementary Table S1 lists all 20,171 pairs with their class, added and removed residues, manifest scores, model chain lengths and high-confidence calls. Table S2 gives the transitions by class under the deposition call, under the ipSAE threshold alone and over all pairs from served models, and Table S3 the logistic models. Table S4 lists the interface retention of the 1,678 pairs with a high-confidence canonical, Table S5 the 1,247 high-confidence isoform homodimers among the 20,171 pairs, with their interface composition, and Table S6 the identical-sequence copies. Table S7 gives the transitions by class under each alternative call, Table S8 the replacement swaps with their alignment to the sequence they replace, and Table S9 the added-sequence confidence of every isoform in the monomer models. Tables S1 to S9 are in the workbook Supplementary_Tables.xlsx, and Table S10 is below.

Supplementary Table S10. Classes of human isoform-canonical pairs and the residues they add and remove.

Class	Pairs	Shorter / equal / longer	Ten or more added residues	Median added residues	Median removed residues
Terminal truncation	2,952	2,952 / 0 / 0	0	0	104
Internal deletion	4,948	4,948 / 0 / 0	0	0	41
Replacement	10,858	9,288 / 115 / 1,455	7,516	19	109
Addition	1,405	0 / 0 / 1,405	1,014	21	0
No shared block	8	8 / 0 / 0	8	87.5	403.5
All	20,171		8,538		

Pairs in which both sequences are 15 to 1,500 residues long. Added residues are isoform residues outside the identical blocks of five or more residues shared with the canonical, and removed residues are canonical residues outside them.